Statistics & Probability

ANSWER KEY

Section 1: Measures of Central Tendency & Dispersion

1 The numbers $4, 7, 9, 12, 18$ have a mean of $\bar{x} = 10$. Find the standard deviation σ correct to 2 decimal places.

a. 4.73 ✓

b. 5.21

c. 4.12

d. 3.85

Squared deviations: $(4-10)^2 = 36, (7-10)^2 = 9, (9-10)^2 = 1, (12-10)^2 = 4, (18-10)^2 = 64$. Sum $= 114$. Variance $= 114/5 = 22.8$. $\sigma = \sqrt{22.8} \approx 4.73$.

2 A data set of 20 observations has $\sum x = 240$ and $\sum x^2 = 3280$. Find the variance of the data set.

a. **20** ✓

b. 16

c. 24

d. 12

Mean $\bar{x} = 240/20 = 12$. Variance $= \frac{\sum x^2}{n} - \bar{x}^2 = \frac{3280}{20} - 12^2 = 164 - 144 = 20$.

3 In a box-and-whisker plot, the lower quartile is $Q_1 = 24$ and the upper quartile is $Q_3 = 42$. According to the $1.5 \times$ IQR rule, what is the boundary for outliers at the upper end?

a. 69 ✓

b. 60

c. 65

d. 72

IQR $= Q_3 - Q_1 = 42 - 24 = 18$. Upper boundary $= Q_3 + 1.5 \times \text{IQR} = 42 + 1.5(18) = 42 + 27 = 69$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Section 2: Probability Rules & Conditional Probability

4 Two independent events A and B have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \cup B)$.

a. 0.70 ✓

b. 0.90

c. 0.20

d. 0.65

For independent events, $P(A \cap B) = P(A)P(B) = 0.4 \times 0.5 = 0.2$. Then $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - 0.2 = 0.70$.

5 Given $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, find the conditional probability $P(A \mid B)$.

a. 0.60 ✓

b. 0.50

c. 0.30

d. 0.75

$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = 0.60$.

6 A bag contains 4 red balls and 6 blue balls. Two balls are drawn at random without replacement. What is the probability that both balls are red?

a. $\frac{2}{15}$ ✓

b. $\frac{4}{25}$

c. $\frac{1}{6}$

d. $\frac{1}{5}$

$P(R_1) = 4/10 = 2/5$. $P(R_2 \mid R_1) = 3/9 = 1/3$. Combined probability $= \frac{2}{5} \times \frac{1}{3} = \frac{2}{15}$.

Section 3: Discrete Probability Distributions

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 A discrete random variable X has probability function $P(X = x) = kx$ for $x \in \{1, 2, 3, 4\}$. Find the value of k.

a. $\frac{1}{10}$ ✓

b. $\frac{1}{6}$

c. $\frac{1}{4}$

d. $\frac{1}{12}$

Sum of probabilities must equal 1: $k(1 + 2 + 3 + 4) = 1 \Rightarrow 10k = 1 \Rightarrow k = 1/10$.

8 Find the expectation $E(X)$ for the distribution where $P(X = 1) = 0.2, P(X = 2) = 0.5, P(X = 3) = 0.3$.

a. 2.1 ✓

b. 2.0

c. 2.3

d. 1.9

$E(X) = \sum xP(X = x) = 1(0.2) + 2(0.5) + 3(0.3) = 0.2 + 1.0 + 0.9 = 2.1$.

9 For a discrete random variable X with $E(X) = 4$ and $E(X^2) = 25$, what is the variance $\text{Var}(X)$?

a. 9 ✓

b. 5

c. 16

d. 21

$\text{Var}(X) = E(X^2) - [E(X)]^2 = 25 - 4^2 = 25 - 16 = 9$.

Section 4: Binomial Distribution & Exam Applications

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 A random variable $X \sim B(8, 0.25)$. Find $P(X = 2)$ correct to 4 decimal places.

a. 0.3115 ✓

b. 0.2076

c. 0.2670

d. 0.1038

$P(X = 2) = \binom{8}{2}(0.25)^2(0.75)^6 = 28 \times 0.0625 \times 0.1779785 \approx 0.3115.$

11 If $X \sim B(100, 0.4)$, find the mean and variance of X.

a. **Mean** $= 40$**, Variance** $= 24$ ✓

b. Mean $= 40$, Variance $= 16$

c. Mean $= 50$, Variance $= 25$

d. Mean $= 40$, Variance $= 4.9$

Mean $\mu = np = 100(0.4) = 40$. Variance $\sigma^2 = np(1-p) = 100(0.4)(0.6) = 24$.

12 A multiple-choice quiz has 5 questions, each with 4 options and exactly one correct answer. If a student guesses at random on all 5 questions, what is the probability of passing with at least 3 correct answers?

a. $\frac{53}{512} \approx 0.1035$ ✓

b. 0.2500

c. 0.0879

d. 0.1563

$P(X \geq 3) = P(3) + P(4) + P(5) = \binom{5}{3}(1/4)^3(3/4)^2 + \binom{5}{4}(1/4)^4(3/4) + \binom{5}{5}(1/4)^5 = (10 \times 9 + 5 \times 3 + 1)/1024 = (90$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647